



Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Operations Research

Exam: IN0024 / Endterm
Examiner: Prof. Dr. Martin Bichler

Date: Friday 7th August, 2026
Time: 17:00 – 19:00

Working instructions

- This exam consists of **20 pages** with a total of **7 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 121 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **non-programmable pocket calculator**
 - one **analog dictionary** English ↔ native language
- Subproblems marked by * can be solved without results of previous subproblems.
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____

Problem 1 Multiple Choice Questions (16 credits)

Mark all correct answers (there can be more than one):

Scoring: Each question is worth up to 2 points. You receive 2 points if you select all correct options and no incorrect ones. Selecting one incorrect option or not selecting a correct option reduces the score to 1 point. Any greater deviation results in 0 points.

Mark correct answers with a cross



To undo a cross, completely fill out the answer option



To re-mark an option, use a human-readable marking



a) Which of the following statements are correct?

- ☐ The simplex algorithm always finds an optimal solution to a linear program in polynomial time.
- ☐ Consider $\min\{c^T x : Ax \leq b, x \in \mathbb{Z}^n\}$. If A is totally unimodular, then the optimal solution can be found in polynomial time.
- ☒ Every linear program can be solved in polynomial time.
- ☐ The branch-and-bound algorithm is guaranteed to find an optimal solution to an integer program in polynomial time.

b) Which of the following statements are correct?

- ☒ If the primal problem is unbounded, then the dual problem is infeasible.
- ☐ If the feasible region is unbounded, then any linear program defined on it has an unbounded objective value.
- ☒ If both the primal and dual problems are feasible, then they both have optimal solutions.
- ☒ If the dual problem is unbounded, then the primal problem is infeasible.

c) Consider a maximization integer linear program solved using the Branch-and-Bound algorithm. Which of the following statements are correct?

- ☒ A node can be pruned if the value of its LP relaxation is worse than the value of the current incumbent solution.
- ☐ Branch-and-Bound always explores every feasible integer solution before proving optimality.
- ☒ For a maximization problem, the objective value of the LP relaxation provides an upper bound on the objective value achievable by any integer-feasible solution in that node.
- ☒ If the LP relaxation at a node has an integer-optimal solution, then that solution is optimal for the corresponding subproblem.

d) Consider a pure integer linear program whose LP relaxation has been solved optimally. The optimal LP solution is fractional. Which of the following statements are correct?

- ☐ A Gomory cut can be generated from the objective row of the simplex tableau.
- ☒ A Gomory cut can be derived from a row of the optimal simplex tableau whose basic variable has a fractional value.
- ☐ If a Gomory cut is added to the LP relaxation, the optimal objective value of the LP relaxation can only improve (increase) for a maximization problem.
- ☒ A Gomory cut removes the current fractional LP-optimal solution but does not remove any feasible integer solution.

e) Which statements about total unimodularity are correct?

- ☒ A matrix containing an entry 2 cannot be totally unimodular.
- ☐ If A is totally unimodular and b is integral, then every feasible point of $\{x : Ax \leq b\}$ is integral.
- ☒ In a totally unimodular matrix, every square submatrix has determinant in $\{-1, 0, 1\}$.
- ☐ Every square matrix with determinant 1 is totally unimodular.

f) A 1.5-approximation algorithm for a metric TSP instance returns a tour of cost 30. Let OPT denote the cost of an optimal tour. Which conclusions are guaranteed?

- ☐ $OPT = 20$.
- ☒ $OPT \leq 30$.
- ☒ $OPT \geq 20$.
- ☐ The returned tour must be optimal.

g) Which of the following statements are correct?

- ☒ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable and $\nabla^2 f(x) \succeq 0$ for all x , then f is convex.
- ☐ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, then every stationary point of f is a local maximum.
- ☒ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable, then every local minimizer of f is also a global minimizer.
- ☐ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly convex, then it always has a minimizer.

h) Which of the following statements are correct?

- ☒ For a convex minimization problem, any feasible point satisfying the KKT conditions is a global minimizer.
- ☒ For a convex minimization problem satisfying Slater's condition, the KKT conditions are necessary for optimality.
- ☐ A KKT point of a non-convex optimization problem is always a global optimum.
- ☐ Slater's condition requires that all feasible points satisfy all inequality constraints strictly.

Additional space for notes:

Problem 2 Glass Production (21 credits)

A manufacturer develops specialty glasses. The glass is produced by mixing different oxides. The composition influences the physical properties of the glass as well as its production cost.

Let M denote the set of available raw materials

$$M = \{\text{SiO}_2, \text{Li}_2\text{O}, \text{Na}_2\text{O}, \text{K}_2\text{O}, \text{MgO}, \text{CaO}, \text{SrO}, \text{BaO}, \text{B}_2\text{O}_3, \text{Al}_2\text{O}_3\}.$$

For every material $i \in M$, $x_i \in [0, 1]$ denotes its mass fraction in the final glass. Furthermore, $y_i \in \{0, 1\}$ indicates whether material i is used. The unit cost of material i is denoted by c_i .

The customer specifies acceptable ranges for several material properties:

$$P = \{\text{conductivity, thermal expansion, refractive index, density, opacity}\}$$

For every property $k \in P$, a linear model predicts the value of property k given the glass composition:

$$P_k(x) = \sum_{i \in M} a_{ki} x_i.$$

Formulate the optimization problem mathematically using only linear constraints.



a) The objective is to minimize production cost.

$$\min \sum_{i \in M} c_i x_i, \quad y_i \geq x_i.$$



b) The total composition must sum to 100%.

$$\sum_{i \in M} x_i = 1$$



c) Silicon dioxide (SiO_2) forms the glass network. For mechanical stability, at least 60% of the glass must consist of SiO_2 .

$$x_{\text{SiO}_2} \geq 0.6$$



d) Every selected material must contribute at least 2%.

$$x_i \geq 0.02 y_i \quad \forall i \in M$$



e) The produced glass needs to satisfy the property specifications defined by lower bounds L_k and upper bounds U_k for all properties.

$$L_k \leq \sum_{i \in M} a_{ki} x_i \leq U_k \quad \forall k \in P$$



f) Due to an existing patent, lithium oxide (Li_2O) and potassium oxide (K_2O) may not be used simultaneously.

$$y_{\text{Li}_2\text{O}} + y_{\text{K}_2\text{O}} \leq 1$$

g) Barium oxide (BaO) increases the refractive index of the glass but reduces its chemical durability. Therefore, barium oxide needs to be combined with aluminum oxide (Al_2O_3), to stabilize the glass network. Consequently, whenever BaO is used, Al_2O_3 must also be present.



$$y_{\text{BaO}} \leq y_{\text{Al}_2\text{O}_3}$$

h) At most six different raw materials may be used.



$$\sum_{i \in M} y_i \leq 6$$

i) Given an integer variable $z \in \{0, 1\}$ and a variable $x \in [0, 1]$ model the product zx linearly using a new variable w .



$$w \leq x, \quad w \leq z, \quad w \geq x + z - 1, \quad w \geq 0$$

j) *Boron anomaly:*

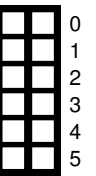
The coefficient of thermal expansion is increasing for small concentrations of boron oxide (B_2O_3), but changes significantly once the concentration exceeds a threshold τ . Consequently, the thermal expansion can no longer be described by a single linear function.

Incorporate this effect into the optimization model.

Hint: introduce a binary variable $z \in \{0, 1\}$ that captures whether the concentration of B_2O_3 is above, or below the threshold τ and enforce this behavior suitably using linear constraints.

Use the z to define the new thermal expansion model $P_{\text{exp}}(x, z)$.

You can use the previous subproblem without proof to eliminate products of z and x_i .



$$z = \begin{cases} 0, & x_{\text{B}_2\text{O}_3} < \tau, \\ 1, & x_{\text{B}_2\text{O}_3} \geq \tau. \end{cases}$$

As $x_i \in [0, 1]$ it follows

$$x_{\text{B}_2\text{O}_3} \leq \tau + z,$$

$$x_{\text{B}_2\text{O}_3} \geq \tau z.$$

The thermal expansion model can then be written as

$$P_{\text{exp}} = \sum_{i \neq \text{B}_2\text{O}_3} a_i x_i + [(1 - z)a_{\text{B}_2\text{O}_3, L} - za_{\text{B}_2\text{O}_3, H}]x_{\text{B}_2\text{O}_3}.$$

Now we introduce a new variable w to eliminate the product of $x_{\text{B}_2\text{O}_3}$ and z as in the previous subproblem.

Problem 3 Duality (15 credits)

Consider the following primal linear program (P):

$$\begin{aligned} \max \quad & 4x_1 + 2x_2 + 3x_3 \\ \text{s.t.} \quad & x_1 + x_2 + 2x_3 \leq 8 \\ & 2x_1 - x_2 + x_3 \geq 2 \\ & x_1 + x_2 = 4 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

a) Derive the dual linear program (D) corresponding to (P).

The dual problem (D) is:

$$\begin{aligned} \min \quad & 8y_1 + 2y_2 + 4y_3 \\ \text{s.t.} \quad & y_1 + 2y_2 + y_3 \geq 4 \\ & y_1 - y_2 + y_3 \geq 2 \\ & 2y_1 + y_2 \geq 3 \\ & y_1 \geq 0, \quad y_2 \leq 0, \quad y_3 \in \mathbb{R}. \end{aligned}$$

b) State the primal and the dual complementary slackness conditions explicitly.

	0
	1
	2
	3
	4

Primal CS conditions:

$$(x_1 + x_2 + 2x_3 - 8)y_1 = 0$$

$$(2x_1 - x_2 + x_3 - 2)y_2 = 0$$

$$(x_1 + x_2 - 4)y_3 = 0$$

Dual CS conditions:

$$(y_1 + 2y_2 + y_3 - 4)x_1 = 0$$

$$(y_1 - y_2 + y_3 - 2)x_2 = 0$$

$$(2y_1 + y_2 - 3)x_3 = 0$$

c) Consider the feasible primal solution

$$x^* = (2, 2, 0)^T$$

and the feasible dual solution

$$y^* = (2, 0, 2)^T.$$

Can these two solutions both be optimal? Justify your answer.

	0
	1
	2
	3
	4

The primal objective value is

$$4 \cdot 2 + 2 \cdot 2 + 3 \cdot 0 = 12.$$

The dual objective value is

$$8 \cdot 2 + 2 \cdot 0 + 4 \cdot 2 = 24.$$

Since the objective values are not equal, from the strong duality theorem it follows that the two feasible solutions cannot both be optimal.

Problem 4 Branch and Bound (15 credits)

We consider the following knapsack problem with free variables:

$$\begin{aligned} \max_x \quad & \sum_i v_i x_i = 20x_1 + 12x_2 + 2x_3 + 9x_4 \\ \text{s.t.} \quad & \sum_i w_i x_i = 2x_1 + 2x_2 + x_3 + 2x_4 \leq c = 5. \end{aligned} \quad (\mathcal{P}_{\text{free}})$$

The above model is denoted by $\mathcal{P}_{\text{free}}$, in which the variables can take any negative or positive real value. We also consider the same problem when the variables are restricted to be non-negative reals, non-negative integers, or binary values, and denote them by $\mathcal{P}_{\geq 0}$, $\mathcal{P}_{\mathbb{Z}_+}$, and $\mathcal{P}_{\{0,1\}}$, respectively. We want to solve these problems using the Branch and Bound algorithm.

a) Are the problems $\mathcal{P}_{\text{free}}$ and $\mathcal{P}_{\geq 0}$ useful in solving the integer valued programs? Give your reasoning.

Problem $\mathcal{P}_{\text{free}}$ is not useful, because variables can take any negative value, we can set x_3 to an arbitrarily large negative number (e.g., $x_3 = -M$). This creates infinite capacity in the constraint ($2x_1 + 2x_2 + 2x_4 \leq 5 + M$), allowing us to make x_1 arbitrarily large. Thus, the objective value is unbounded ($+\infty$). An unbounded relaxation provides no finite upper bound to prune branches in a Branch and Bound tree.

Problem $\mathcal{P}_{\geq 0}$ is useful, as it is the standard LP relaxation of $\mathcal{P}_{\mathbb{Z}_+}$. Therefore, its optimal objective value provides a valid, finite upper bound for the integer program, which is essential for bounding and pruning.

b) Apply the Branch and Bound algorithm for solving $\mathcal{P}_{\mathbb{Z}_+}$. Only iterate the first branch and solve the resulting subproblems. Also explain how the algorithm would further branch out without solving the resulting subproblems. You do not need to find the optimal solution. *Hint: The LP relaxation of $\mathcal{P}_{\mathbb{Z}_+}$ has the following optimal solution, which you may use without justification: Determine the value-weight ratios $q_i = v_i/w_i$ and index the items such that $q_j \geq q_{j+1}$ for all $j = 1, \dots, n-1$. Set $x_k = c/w_k$ for $k = 1$, and $x_j = 0$ for $j > 1$. When further constraints are imposed, the optimal solution greedily takes items in decreasing order of their value-to-weight ratio until the capacity is full.*

Node 0 (Root): Solved using the LP relaxation, all capacity goes to Item 1.

- Solution: $x_1 = 5/2 = 2.5$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$.
- Bound: $z_0 = 20(2.5) = 50$.
- Branch on x_1 : Branch 1 ($x_1 \leq 2$) and Branch 2 ($x_1 \geq 3$).

Node 1 ($x_1 \leq 2$): Fix $x_1 = 2$. Remaining capacity $c' = 5 - 2(2) = 1$. The next best available item is Item 2 ($w_2 = 2$). We fractionally fill it: $x_2 = 1/2 = 0.5$.

- Solution: $x_1 = 2$, $x_2 = 0.5$, $x_3 = 0$, $x_4 = 0$.
- Bound: $z_1 = 20(2) + 12(0.5) = 46$.
- Branch on x_2 : Node 3 ($x_2 \leq 0$) and Node 4 ($x_2 \geq 1$).

Node 2 ($x_1 \geq 3$): Requires weight $\geq 3 \times 2 = 6$. Capacity is only 5. Pruned by Infeasibility.

	0
	1
	2
	3
	4
	5
	6
	7

c) Solve $\mathcal{P}_{\{0,1\}}$ using the Branch and Bound algorithm. *Hint: The LP relaxation of $\mathcal{P}_{\{0,1\}}$ has the following optimal solution, which you may use without justification: Determine the value-weight ratios $q_i = v_i/w_i$ and index the items such that $q_i \geq q_{i+1}$ for all $j = 1, \dots, n-1$. Choose k such that $\sum_{j=1}^k w_j \leq c$, but $\sum_{j=1}^{k+1} w_j > c$. Set $x_j = 1$ for $j = 1, \dots, k$, $x_{k+1} = (c - \sum_{j=1}^k w_j)/w_{k+1}$, and $x_j = 0$ for $j > k+1$. When further constraints are imposed, the optimal solution greedily takes items in decreasing order of their value-to-weight ratio until the capacity is full.*

The value-weight ratios are 10, 6, 2, 4.5 for x_i with respect to the order $i = 1, 2, 3, 4$. Keeping the same index i , the variables are ordered as x_1, x_2, x_4, x_3 so that these ratios are in decreasing order.

Node 0 (Root):

- Solution: $x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 0.5$.
- Bound: $z_0 = 20(1) + 12(1) + 9(0.5) = 36.5$.
- Branch on x_4 : Node 1 ($x_4 = 0$) and Node 2 ($x_4 = 1$).

Node 1 ($x_4 = 0$):

- Solution: $x_1 = 1, x_2 = 1, x_3 = 1, x_4 = 0$.
- Objective: $z_1 = 20(1) + 12(1) + 2(1) = 34$.
- Pruned by Integrality. Incumbent found: $z^* = 34$.

Node 2 ($x_4 = 1$):

- Solution: $x_1 = 1, x_2 = 0.5, x_3 = 0, x_4 = 1$.
- Bound: $z_2 = 20(1) + 12(0.5) + 9(1) = 35$.
- Branch on x_2 : Node 3 ($x_2 = 0$) and Node 4 ($x_2 = 1$).

Node 3 ($x_4 = 1, x_2 = 0$):

- Solution: $x_1 = 1, x_2 = 0, x_3 = 1, x_4 = 1$.
- Objective: $z_3 = 20(1) + 2(1) + 9(1) = 31$.
- Pruned by Integrality/Bound ($31 < z^* = 34$).

Node 4 ($x_4 = 1, x_2 = 1$):

- Solution: $x_1 = 0.5, x_2 = 1, x_3 = 0, x_4 = 1$.
- Bound: $z_4 = 20(0.5) + 12(1) + 9(1) = 31$.
- Pruned by Bound: ($31 < z^* = 34$).

Optimal Solution for $\mathcal{P}_{\{0,1\}}$: $x^* = (1, 1, 1, 0)$ with $z^* = 34$.

Problem 5 Gomory Cut (15 credits)

Consider the following integer programme:

$$\max z = 5x_1 + 8x_2 \quad (5.1)$$

$$\text{s.t. } x_1 + x_2 + s_1 = 6 \quad (5.2)$$

$$5x_1 + 9x_2 + s_2 = 45 \quad (5.3)$$

$$s_1, s_2 \geq 0 \quad (5.4)$$

$$x_1, x_2 \in \mathbb{N}_0 \quad (5.5)$$

After solving the LP relaxation with the simplex method, the following optimal tableau is obtained:

	x_1	x_2	s_1	s_2	RHS
z	0	0	$5/4$	$3/4$	$165/4$
x_1	1	0	$9/4$	$-1/4$	$9/4$
x_2	0	1	$-5/4$	$1/4$	$15/4$

a) Determine the LP-optimal solution and explain why it is not feasible for the integer program.

From the tableau rows we obtain

$$x_1 = \frac{9}{4}, \quad x_2 = \frac{15}{4}.$$

Since neither $x_1 = \frac{9}{4}$ nor $x_2 = \frac{15}{4}$ is an integer, the LP-optimal solution is not feasible for the integer program.

b) Derive a Gomory fractional cut from the row corresponding to the basic variable x_2 . Show all intermediate calculations.

The tableau row is

$$x_2 - \frac{5}{4}s_1 + \frac{1}{4}s_2 = \frac{15}{4}.$$

We round down and move the fractional parts to the right-hand side.

$$x_2 - 2s_1 - 3 = \frac{3}{4} - \frac{3}{4}s_1 - \frac{1}{4}s_2.$$

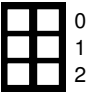
The Gomory fractional cut is therefore

$$\frac{3}{4}s_1 + \frac{1}{4}s_2 \geq \frac{3}{4} \quad \text{or} \quad x_2 - 2s_1 \leq 3$$

Multiplying the first one by 4 gives the equivalent form

$$3s_1 + s_2 \geq 3.$$

c) Verify that the current LP-optimal solution violates the Gomory cut obtained in the previous part.



At the current LP-optimal solution,

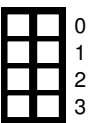
$$s_1 = 0, \quad s_2 = 0.$$

Substituting into the cut yields

$$3(0) + 0 = 0.$$

However, $0 \not\geq 3$. Hence the LP-optimal solution violates the Gomory cut.
Similarly, $x_2 = \frac{15}{4}$ and $s_1 = 0$ violates the other Gomory cut.

d) Formulate the LP relaxation that would be solved after adding the cut obtained in Part (2). You may reference original constraints (1)–(5) instead of rewriting them completely.



The next LP relaxation is obtained by adding the Gomory cut to the original LP relaxation.

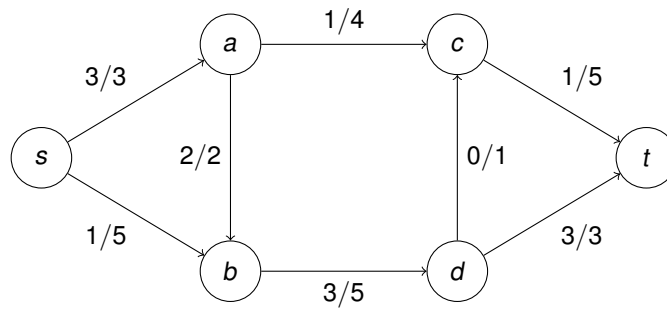
$$\begin{aligned} &\max (1) \\ &\text{s.t. } (2), (3), \\ &\quad 3s_1 + s_2 \geq 3 \\ &\quad x_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

This LP would be solved in the next iteration of the Gomory cutting-plane algorithm.
Adding the other cut yields:

$$\begin{aligned} &\max (1) \\ &\text{s.t. } (2), (3), \\ &\quad x_2 - 2s_1 \leq 3 \\ &\quad x_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

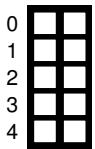
Problem 6 Maximum Flow (17 credits)

Consider the following flow network. The label f/u on an arc denotes its current flow f and capacity u .



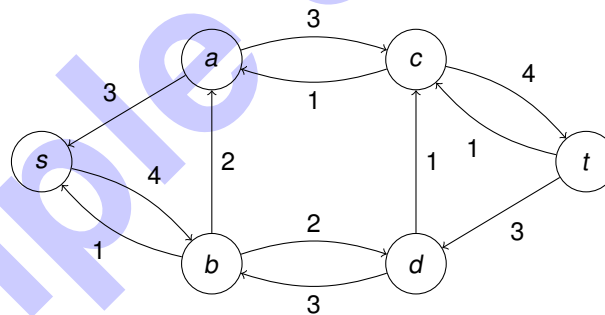
a) State the value of the current flow.

The value of the current flow is $3 + 1 = 4$.



b) Determine all arcs with positive capacity in the residual network and state their capacities. You may present your answer as a drawing or a list.

The residual network is:



	0
	1
	2
	3
	4
	5
	6
	7

c) Starting from the given flow, run the Ford–Fulkerson algorithm until no augmenting s - t path remains. For every augmentation, state:

- the chosen s - t path in the current residual network;
- the amount by which the flow is augmented; and
- the resulting flow on every original arc whose flow changes.

Finally, state the value of the resulting flow.

One valid order of augmentations is the following.

First use the residual path (s, b, d, c, t) . Its residual capacities are 4, 2, 1, 4, so the augmentation amount is 1. The changed original flows are

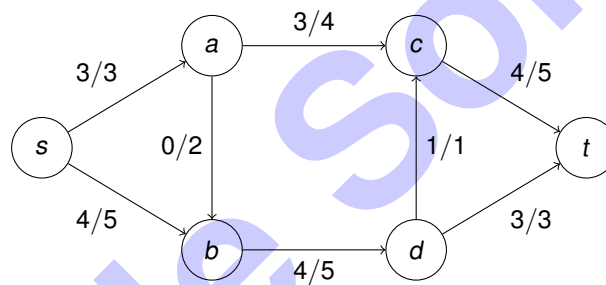
$$f(s, b) : 1 \rightarrow 2, \quad f(b, d) : 3 \rightarrow 4, \quad f(d, c) : 0 \rightarrow 1, \quad f(c, t) : 1 \rightarrow 2.$$

The flow value is now 5.

Next use the residual path (s, b, a, c, t) . Its residual capacities are 3, 2, 3, 3, so the augmentation amount is 2. Since (b, a) is a backward residual arc, the flow on (a, b) is decreased. The changed original flows are

$$f(s, b) : 2 \rightarrow 4, \quad f(a, b) : 2 \rightarrow 0, \quad f(a, c) : 1 \rightarrow 3, \quad f(c, t) : 2 \rightarrow 4.$$

For reference, the resulting flow is shown below; the labels are again of the form f/u .



Its value is $3 + 4 = 7$. There is no residual s - t path.

The two augmentations may also be performed in the reverse order. Their augmentation amounts are again 2 and 1, respectively, and they yield the same final flow.

0		
1		
2		
3		

d) Find an s - t cut whose capacity equals the value of the resulting flow. State the two node sets defining the cut and compute its capacity. Explain why this proves that the resulting flow is maximum.

Consider the cut

$$(\{s, b, d\}, \{a, c, t\}).$$

Its capacity is

$$u(s, a) + u(d, c) + u(d, t) = 3 + 1 + 3 = 7.$$

The flow value equals the capacity of this cut. By the maximum-flow minimum-cut theorem, the flow is maximum and the cut is minimum.

0		
1		
2		

e) Briefly justify why a maximum flow can be chosen integral when all capacities are integral.

Ford–Fulkerson can start with the integral zero flow. If all capacities are integral, then all residual capacities and every augmentation amount are integral. Thus, every augmentation preserves integrality, and the algorithm returns an integral maximum flow when it terminates.

Alternatively, the constraint matrix of the maximum-flow LP is totally unimodular: it consists of the node–arc incidence matrix together with identity rows for the capacity constraints. Since the right-hand side is integral, the LP has an integral optimal vertex. Hence, a maximum flow can be chosen integral.

Problem 7 Budget Allocation (22 credits)

A company allocates one unit of budget between two advertising channels. Let $x_1 \in [0, 1]$ denote the fraction of the budget assigned to channel 1 and $x_2 \in [0, 1]$ the fraction assigned to channel 2. Due to diminishing returns, the resulting utility is modeled as

$$f_\alpha(x) = \log(1 + x_1) + \alpha \log(1 + x_2),$$

where $\alpha > 0$ describes the relative importance of channel 2 and $\log$ denotes the natural logarithm. The company wants to solve

$$(P_\alpha) \quad \max_{x \in \mathbb{R}^2} \log(1 + x_1) + \alpha \log(1 + x_2)$$

subject to

$$x_1 + x_2 \leq 1, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

Equivalently, the inequality constraints can be written as

$$g_1(x) = x_1 + x_2 - 1 \leq 0, \quad g_2(x) = -x_1 \leq 0, \quad g_3(x) = -x_2 \leq 0.$$

a) State Slater's condition for a convex optimization problem with inequality constraints.

For a convex optimization problem

$$\min f(x) \quad \text{s.t.} \quad g_i(x) \leq 0, \quad h_j(x) = 0,$$

with f, g_i convex and h_j affine, Slater's condition requires the existence of $\bar{x}$ such that

$$g_i(\bar{x}) < 0 \quad \forall i, \quad h_j(\bar{x}) = 0 \quad \forall j.$$



b) Show that f_α is concave on the feasible set.

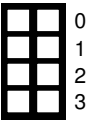
We compute

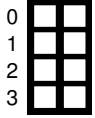
$$\nabla^2 f_\alpha(x) = \begin{pmatrix} -\frac{1}{(1+x_1)^2} & 0 \\ 0 & -\frac{\alpha}{(1+x_2)^2} \end{pmatrix}.$$

Since $\alpha > 0$ and $x_1, x_2 \geq 0$, both diagonal entries are non-positive, and the matrix is diagonal, so all eigenvalues are non-positive:

$$\nabla^2 f_\alpha(x) \preceq 0.$$

Therefore f_α is concave on the feasible set.





c) Show that the feasible set of (P_α) is convex and that Slater's condition holds.

The feasible set is

$$X = \{x \in \mathbb{R}^2 \mid x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0\}.$$

It is the intersection of the three halfspaces

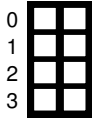
$$x_1 + x_2 \leq 1, \quad -x_1 \leq 0, \quad -x_2 \leq 0,$$

and an intersection of convex sets is convex, hence X is convex.

Moreover, $\bar{x} = (\frac{1}{3}, \frac{1}{3})^T$ is strictly feasible, since

$$\frac{1}{3} + \frac{1}{3} < 1, \quad \frac{1}{3} > 0, \quad \frac{1}{3} > 0,$$

so Slater's condition holds. *Any strictly feasible point is accepted.*



d) In addition to the utility, the company accounts for an administrative overhead proportional to the larger of the two allocations,

$$c(x) = \max\{x_1, x_2\}.$$

State the definition of a convex function and show *directly from the definition* that c is convex on $\mathbb{R}^2$.

A function $c : \mathbb{R}^n \rightarrow \mathbb{R}$ with convex domain is convex if

$$c(\theta x + (1 - \theta)y) \leq \theta c(x) + (1 - \theta)c(y) \quad \forall x, y \in \mathbb{R}^n, \forall \theta \in [0, 1].$$

The domain $\mathbb{R}^2$ is convex.

Let $x, y \in \mathbb{R}^2$ and $\theta \in [0, 1]$. For each $i \in \{1, 2\}$ we have $x_i \leq c(x)$ and $y_i \leq c(y)$, hence, using $\theta \geq 0$ and $1 - \theta \geq 0$,

$$\theta x_i + (1 - \theta)y_i \leq \theta c(x) + (1 - \theta)c(y).$$

The right-hand side does not depend on i , so taking the maximum over $i \in \{1, 2\}$ on the left gives

$$c(\theta x + (1 - \theta)y) = \max_i \{\theta x_i + (1 - \theta)y_i\} \leq \theta c(x) + (1 - \theta)c(y).$$

Consequently $-\beta c$ is concave for $\beta \geq 0$, and as a sum of concave functions $f_\alpha - \beta c$ is concave.

In the following subproblems, set up the KKT conditions of (P_α) and determine the optimal solution x^* for arbitrary $\alpha > 0$, distinguishing cases for α . Work through the following steps.

e) Write (P_α) as a minimization problem, state the Lagrangian, and write down all KKT conditions.

	0
	1
	2
	3

We solve the equivalent problem $\min -f_\alpha(x)$. The Lagrangian is

$$L(x, \lambda) = -\log(1 + x_1) - \alpha \log(1 + x_2) + \lambda_1(x_1 + x_2 - 1) - \lambda_2 x_1 - \lambda_3 x_2.$$

The KKT conditions are

$$\begin{aligned} -\frac{1}{1+x_1} + \lambda_1 - \lambda_2 &= 0, & -\frac{\alpha}{1+x_2} + \lambda_1 - \lambda_3 &= 0, \\ x_1 + x_2 &\leq 1, & x_1 &\geq 0, & x_2 &\geq 0, & \lambda_1, \lambda_2, \lambda_3 &\geq 0, \\ \lambda_1(x_1 + x_2 - 1) &= 0, & \lambda_2 x_1 &= 0, & \lambda_3 x_2 &= 0. \end{aligned}$$

f) Show that the budget constraint is active at every KKT point, i.e. $x_1 + x_2 = 1$.

	0
	1
	2

From the first stationarity condition,

$$\lambda_1 = \frac{1}{1+x_1} + \lambda_2 > 0,$$

since $\frac{1}{1+x_1} > 0$ and $\lambda_2 \geq 0$. Complementary slackness $\lambda_1(x_1 + x_2 - 1) = 0$ then forces

$$x_1 + x_2 = 1.$$

The same argument with the second stationarity condition is equally valid.

0	
1	
2	
3	
4	
5	
6	

g) Using $x_2 = 1 - x_1$ from (f), analyse the three cases $x_1, x_2 > 0$, $x^* = (1, 0)^T$, and $x^* = (0, 1)^T$. For each case determine the multipliers and the range of α for which the case is consistent with the KKT conditions. Summarize the result and justify why the KKT points are global maxima.

Case 1: $x_1 > 0$, $x_2 > 0$. Then $\lambda_2 = \lambda_3 = 0$ and stationarity gives

$$\frac{1}{1+x_1} = \lambda_1 = \frac{\alpha}{1+x_2} = \frac{\alpha}{2-x_1} \Rightarrow 2-x_1 = \alpha(1+x_1),$$

hence

$$x_1^* = \frac{2-\alpha}{1+\alpha}, \quad x_2^* = \frac{2\alpha-1}{1+\alpha}, \quad \lambda_1 = \frac{1+\alpha}{3},$$

and both components are positive iff $\frac{1}{2} < \alpha < 2$.

Case 2: $x^* = (1, 0)^T$. Here $\lambda_2 = 0$ and $\lambda_1 = \frac{1}{1+1} = \frac{1}{2}$, so

$$\lambda_3 = \lambda_1 - \frac{\alpha}{1+0} = \frac{1}{2} - \alpha \geq 0 \iff \alpha \leq \frac{1}{2}.$$

Case 3: $x^* = (0, 1)^T$. Here $\lambda_3 = 0$ and $\lambda_1 = \frac{\alpha}{1+1} = \frac{\alpha}{2}$, so

$$\lambda_2 = \lambda_1 - \frac{1}{1+0} = \frac{\alpha}{2} - 1 \geq 0 \iff \alpha \geq 2.$$

Summary:

$$x^* = \begin{cases} (1, 0)^T, & 0 < \alpha \leq \frac{1}{2}, \\ \left(\frac{2-\alpha}{1+\alpha}, \frac{2\alpha-1}{1+\alpha} \right)^T, & \frac{1}{2} < \alpha < 2, \\ (0, 1)^T, & \alpha \geq 2. \end{cases}$$

Since (P_α) maximizes a concave function over a convex feasible set and Slater's condition holds, the KKT conditions are necessary and sufficient, so these points are global maxima.

Deriving the interior solution by eliminating x_2 and maximizing $\phi(x_1) = \log(1+x_1) + \alpha \log(2-x_1)$ is accepted, provided the boundary cases are still discussed.

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Sample Solution

Sample Solution